

6.1 Populations grow and decline

This chapter will explain:

- ★ the patterns and trends in global population growth and the rapid increase in the world's population
- ★ the reasons for the growth and decline of a country's population
- ★ the impacts of pro-natalist and anti-natalist policies on birth rates
- ★ the strengths and limitations of the demographic transition model.

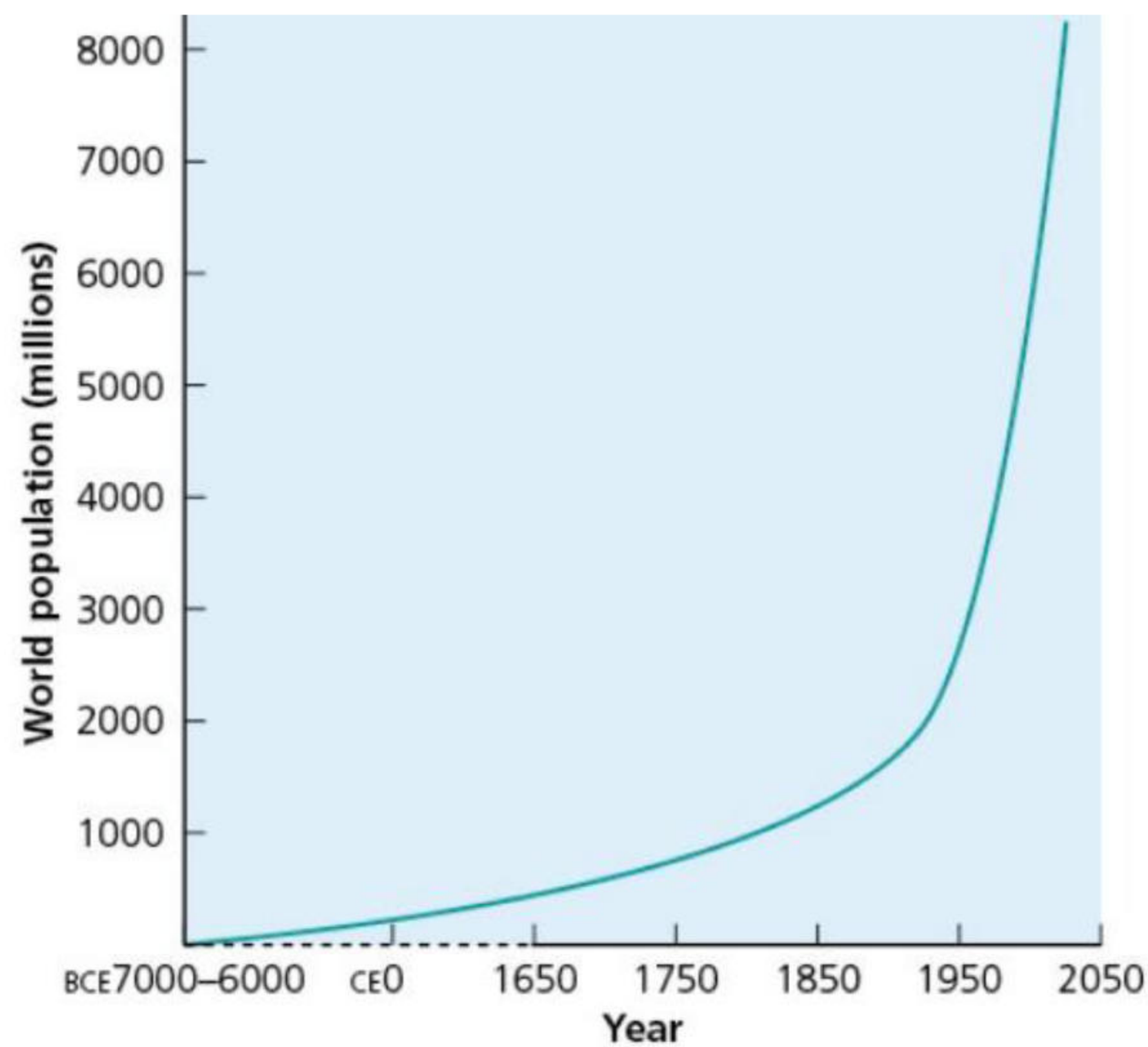
The patterns and trends in global population growth: the rapid increase in the world's population

During most of the period since humankind first evolved global population was very small, reaching perhaps some 125,000 people a million years ago.

- » It has been estimated that 10,000 years ago, when people first began to domesticate animals and cultivate crops, world population was no more than 5 million. Known as the Neolithic Revolution, this period of economic change significantly altered the relationship between people and their environments.
- » As a result of technological advances, the **carrying capacity** of populated areas improved and population increased. By 3500 BCE global population had reached 30 million, and by 0 CE this had risen to about 250 million ([Figure 6.1](#)).
- » **Demographers** estimate that world population reached 500 million by about 1650. From this time, population grew at an increased rate ([Figure 6.2](#)).



▲ **Figure 6.1** The Colosseum, Rome – construction began around 70 CE, when the world's total population was about 250 million



▲ **Figure 6.2** World population growth

By about 1800, global population had doubled to reach 1 billion. [Table 6.1](#) shows the time taken for each subsequent billion to be reached. It took less than 50 years for world population to double from 4 billion in 1974 to 8 billion in 2022. Global population is projected to surpass 9 billion around 2038 and 10 billion by 2060. These figures are based on the UN's medium projection variant. As with all **population projections**, there is a degree of uncertainty because so many different factors will affect future population growth.

▼ **Table 6.1** World population growth by each billion

Population	Year	Number of years to add each billion
1 billion	1804	All of human history
2 billion	1927	123
3 billion	1960	33
4 billion	1974	14
5 billion	1987	13
6 billion	1998	11
7 billion	2010	12
8 billion	2022	12
9 billion	2038	16
10 billion	2060	22

The rapid growth in human population in recent centuries has been due to significant advances in:

- » public health and medicine, including greater access to clean drinking water, improvements in sanitation and advances in disease control
- » nutrition and general living standards.

These massively important improvements in human welfare have lowered considerably the risk of dying, particularly among children (**child mortality rate**). Thus, increasing numbers of people have survived to reproductive age, alongside the gradual increase in human lifespan.

Recent demographic change

Fastest in Africa; slowest in Europe

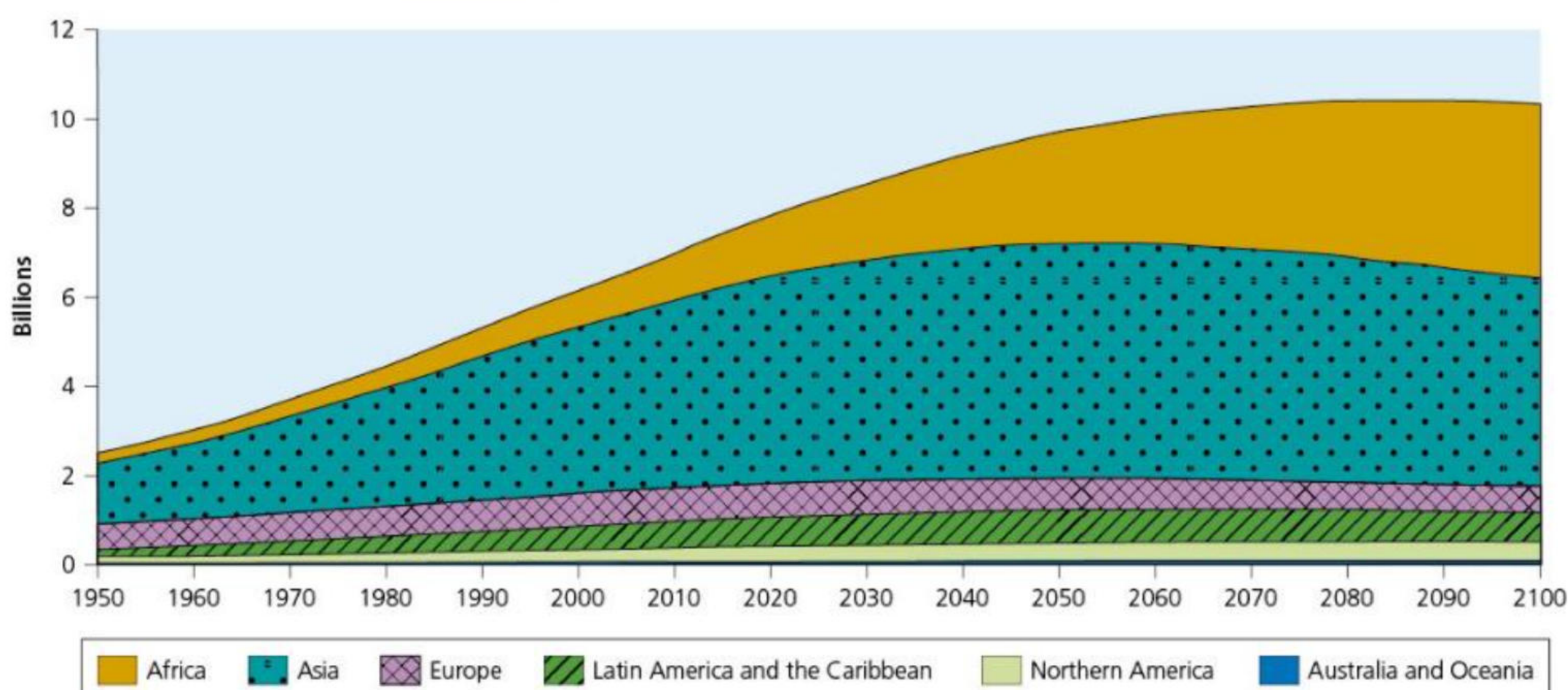
[Figure 6.3](#) shows population growth and projected population growth by world region from 1950 to 2100, while [Table 6.2](#) gives the percentage share of world population by world region for 1980, 2010, 2023 and 2050.

- » Asia dominates global population, with 59.1 per cent of the world's population in 2023.

- » Africa significantly increased its share of global population between 1980 and 2023 and overtook Europe as the second most populous world region in 1996.
- » Together, Asia and Africa accounted for 77.2 per cent of the world's population in 2023.

Asia and Africa are projected to drive much of the world's future population growth, with the fastest rate of growth still being in Africa. This continent is projected to have more than a quarter of the world's population by 2050. By that year, it is expected that just over 80 per cent of the world's population will live in either Asia or Africa.

Europe's share of world population has been in decline for a long time, falling from 22.0 per cent in 1950 to 9.2 per cent in 2023. It is projected to decline further, to 7.3 per cent in 2050. Europe's total population, in absolute terms, began to decline in 2021. Northern America is the other world region where the population share has declined significantly, but at a slower rate than that of Europe. However, in contrast to Europe, there is no indication that Northern America's population is ever going to decline in absolute terms in the time period illustrated by [Figure 6.3](#).



▲ **Figure 6.3** Population change by world region, 1950–2100 (projected)

▼ **Table 6.2** Global population distribution by world region

Region	1980	2010	2023	2050
Africa	10.8	15.1	18.1	25.6
Asia	59.3	60.4	59.1	54.5
Latin America/Caribbean	8.2	8.5	8.3	7.7
North America	5.6	4.9	4.7	4.3
Europe	15.6	10.6	9.2	7.3
Australia/Oceania	0.5	0.5	0.6	0.6

Population momentum

World population data for the period 1950–2023 show that the rate of population growth was much higher in LICs and MICs than in HICs. However, it is only since the mid-twentieth century that population growth in LICs and MICs has overtaken that in HICs. HICs had their period of high population growth in the nineteenth and early twentieth centuries, while for LICs and MICs high population growth has occurred since about 1950. Countries with high levels of fertility tend to be those with relatively low incomes per capita. Over time the growth of the world's population has become increasingly concentrated among the world's lowest-income countries.

The highest ever global population growth rate was reached in the early to mid-1960s, when population growth in LICs and MICs peaked at 2.4 per cent per year. At this time the term 'population explosion' was widely used to describe this rapid population growth, but by the late 1990s the rate of population growth was down to 1.8 per cent per year. However, even though the rate of growth had fallen, **population momentum** meant that the numbers being added each year did not peak until the late 1980s. By 2023, global population growth had declined to 0.9 per cent per year.

The demographic transformation, which took a century to complete in HICs, has occurred in a generation in some LICs and MICs. Fertility has dropped further and faster than most demographers foresaw 30 years ago. To a certain extent Africa has been an exception, where in just over half of countries families of at least four children are the norm and with population growth of 2.4 per cent (2023). However, population

growth rates in Africa have fallen significantly in recent decades, following the trend in other world regions.

The present to 2050

Between now and 2050, almost all of the increase in numbers of children and adults under 65 will occur in low- and lower-middle-income countries. In high- and upper-middle-income countries, the population under 65 will decline, with future growth occurring entirely among the population aged 65 years or over.

Table 6.3 shows the ten most populous countries in the world for 2023 and 2050. In mid-2023, India (Figure 6.4) overtook China as the world’s most populous country. Both nations now have populations over 1.4 billion, meaning each represents nearly 18 per cent of the world’s population.

▼ **Table 6.3** The world’s ten largest countries in population, 2023 and 2050

Country	2023 population (millions)	Country	2050 population (millions)
India	1428.6	India	1670.5
China	1411.3	China	1303.5
USA	335.0	USA	383.6
Indonesia	278.7	Nigeria	377.5
Pakistan	240.5	Pakistan	367.8
Nigeria	223.8	Indonesia	328.9
Brazil	204.0	Brazil	230.8
Bangladesh	173.5	Bangladesh	223.5
Russia	146.9	DR Congo	217.5
Mexico	131.0	Ethiopia	214.8



▲ **Figure 6.4** Traffic congestion in Jaipur, northwest India

Interesting note

The Population Reference Bureau estimates that throughout the history of human population, about 108 billion people have lived on Earth. This means that about 6.5 per cent of all people ever born are alive today.

Activities

- 1 With the help of Figure 6.2 (page 124) and Table 6.1 (page 124), briefly describe the growth of the human population over time.
- 2 Define the terms:
 - a Carrying capacity
 - b Demographer.
- 3 Study Table 6.2 (page 126). Describe the changes and projected changes in global population distribution by world region.